

Course: B. Tech

Branch : Electrical Engineering and Allied

Semester : IV

Subject Code & Name: Electrical Machine-II [BTEEC403]

Max Marks: 60

Date: 24/12/2024

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per COE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q.1 Objective type questions. (Compulsory Question)			12
1	Which of the following motors is non self-starting? a. D.C. series motor b. Synchronous motor c. Squirrel cage induction motor d. Wound round induction motor	Understand	1
2	Which of the following losses, in a synchronous motor, does not vary with load ? a. Windage loss b. Copper losses c. Any of the above d. None of the above	Remember	1
3	An asynchronous motor which works on a leading power factor and does not drive a mechanical load is called as a. static condenser b. condenser c. Synchronous condenser d. none of the above	Understand	1
4	Which of the following single-phase motors has relatively poor starting torque ? a Universal motor b. Repulsion motor c. Capacitor motor d. All single phase motors have zero starting torque	Understand	1
5	Electrical faults in a motor occur most commonly in its a. motor winding b. commutator c. control equipment d. brushes	Remember	1
6	Single phase induction motor usually operates on a. 0.6 power factor lagging b. 0.8 power factor lagging c. 0.8 power factor leading d. unity power factor	Remember	1
7	In an alternator, the frequency of voltage generated depends on a. rotative speed only b. number of poles only c. both (a) and (b) d. none of the above	Understand	1
8	A reluctance motor a. is self-starting b. is constant speed motor c. needs no D.C. d. all of the above	Understand	1

	excitation		
9	If the driving force of both the alternators running in parallel is changed. This will result in change in a. generated voltage b. frequency c. back e.m.f. d. all of the above	Understand	1
10	Which motor is normally free from mechanical and magnetic vibrations ? a. Split phase motor b. Universal motor c. Hysteresis motor d. Shaded pole motor	Remember	1
11	In order to reduce the harmonics in the e.m.f. generated in an alternator a. winding is well distributed b. slots are skewed c. salient pole tips are chamfered d. all of the above	Remember	1
12	Regulation of an alternator supplying leading load is a. always negative b. always positive c. sometimes positive and sometimes negative depending upon load and power factor d. All of the above	Remember	1
Q.2	Solve the following.		12
A)	State and Explain Right Hand Thumb Rule and Fleming's Right Hand Rule.	Remember	6
B)	What is Armature Winding? Define the following terms with diagram (1) Coil Span Factor (2) Distribution Factor (3) Winding factor	Understand	6
Q.3	Solve the following.		12
A)	What is Armature Reaction in Synchronous Machine? Explain Armature Reaction at different power factors with Phasor Diagram	Understand	6
B)	Find the ratios $[T_{st}] / [T_{fl}]$, $[T_{st}] / [T_{mp}]$ and $[T_{fl}] / [T_m]$ in terms of slip.	Analyze	6
Q.4	Solve Any Two of the following.		12
A)	Write Working principle and Derive the EMF equation of synchronous generator	Analyze	6
B)	What is RMF? Find RMF by phasor method	Apply	6

- C) A 3 phase star connected alternator is rated at 1600 KVA 13500 v. The armature resistance and synchronous reactance are 2Ω and 25Ω per phase. Calculate the percentage voltage regulation for a load of 1250 Kw at 0.7 lagging power factor. Evaluate 6

Q.5 Solve Any Two of the following. 12

- A) Compare squirrel cage rotor and slipring rotor type Induction motor Understand 6
- B) Sketch and explain torque-slip characteristics and torque-speed characteristics of three phase induction motor. Analyze 6
- C) What is blocked rotor test of Induction motor Understand 6

Q.6 Solve Any Two of the following. 12

- A) Explain capacitor start motor and the capacitor start capacitor run induction motor with neat circuit diagrams Understand 6
- B) Explain AC servo motors with circuit diagram and torque speed characteristics Remember 6
- C) Explain permanent magnet ac motors. Remember 6

*** End ***